

Programme	B. Sc. Mathematics Honours			
Course Code	MAT6EJ303 (2)			
Course Title	MACHINE LEARNING - I			
Type of Course	Elective (Specialisation- Data Science)			
Semester	V			
Academic Level	300 - 399			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Linear Algebra			
Course Summary	The course develops the basic theory of linear discriminative and generative learning models and techniques for linear regression and classification. Understanding both classical methods and modern neural network approaches will prepare students to tackle a wide range of data science challenges.			

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Describe various regression and classification methods and apply them for simple problems.	Ap	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO2	Apply methods of Bayesian inference to learning problems and analyse the solutions	An	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	Describe the functioning of feedforward neural network models of learning.	U	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book		Pattern Recognition and Machine Learning - Christopher M. Bishop - Springer -2006		
Module	Unit	Content	Hrs (48+12)	Ext. Marks (70)
I	Introduction to Statistical Learning		12	Min 15
	1	Review of probability theory, density and distribution functions		
	2	expectation and covariance, Bayesian probabilities.		
	3	Gaussian distribution: conditional and marginal distributions		
	4	Maximum Likelihood and Bayesian inference for Gaussian		
	5	Decision Theory - inference and decision, loss functions		
	6	Entropy, relative entropy and mutual information		
	Chapter 1 and Section 3 of Chapter 2 from text book.			
II	Linear Regression		12	Min 15
	7	Maximum likelihood and least squares		
	8	Regularized least squares		
	9	Bias-Variance Decomposition		
	10	Bayesian Linear Regression		
	11	Parameter and Predictive Distributions		
	12	Bayesian model comparison		
	Chapter 3 of text book			
III	Linear Classification		12	Min 15
	13	Discriminant functions		
	14	Least squares, Fischer discriminant and the relation between them.		
	15	The perceptron algorithm		
	16	Maximum likelihood classifier		
	17	Probabilistic generative models and Logistic Regression		
	18	Bayesian logistic regression		
	Chapter 4 of text book			

IV	Neural Networks		12	Min 15
	19	Feed forward neural networks		
	20	Network training and gradient descent optimization		
	21	Analysis of error backpropagation		
	22	Hessian matrix and diagonal approximation		
	23	Regularization in neural networks.		
	Chapter 5 of text book			
V	Open Ended		12	
		Model Selection and Validation Non-Uniform Learnability The Run Time of Learning		
References: 1)Understanding Machine Learning From Theory to Algorithms - Shai Shalev Shwartz, Shai Ben David - Cambridge University Press - ISBN 978-1-107-05713-5 - 2014 2) Foundations of Machine Learning - Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar - The MIT Press - 2012				

Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	3	2	2	3	2	3	3	3	1	3
CO 2	3	3	2	2	3	2	3	3	3	1	3
CO 3	3	2	2	2	3	2	3	3	3	1	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓